

Lab Assignment 4: Non-Ideal Operational Amplifier Behavior

Revision: February 1, 2023

Summary

In this lab, you will be exposed to the practical aspects of the operational amplifier (op-amp) that impose limitations on its behavior.

Learning Outcomes: After completing this lab, you should be able to:

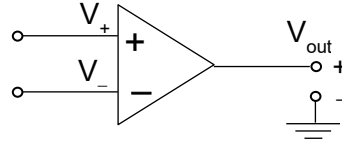
- Understand and obtain key information about the OP27 from its data sheet.
- Design op-amp amplifiers with specific gains and determine their bandwidth based on the constant gain-bandwidth product.
- Experimentally determine the slew rate of the OP27 op-amp and compare it with its typical datasheet value.
- Experimentally determine the input offset voltage of the OP27 and compare it with its typical datasheet value.
- Experimentally determine the input offset and bias currents of the OP27 op-amp and compare them with the datasheet values.
- Experimentally determine the differential open loop gain (A_d) and the common mode gain (A_{cm}) to determine the common mode rejection ratio (CMRR) for the OP27 op-amp and compare it with the datasheet values.

Required Equipment

- EE352 analog parts kit
- Breadboard
- Function generator, oscilloscope
- DMMs
- DC power supplies
- OP27 op-amp

I. The effects of the open-loop gain bandwidth product on op-amp circuits

The open loop op-amp is shown in the figure below.



The open loop gain of a practical op-amp has a finite gain that depends on frequency, where it acts as a first order system with a finite open loop dc gain (A_o) and a low cutoff 3dB frequency (f_b). Hence, the open loop gain can be expressed as

$$A(jf) = \left(\frac{V_{out}}{V_+ - V_-} \right) = \frac{A_o}{1 + j \frac{f}{f_b}} \quad (1)$$

By manipulating (1), we can express it as

$$A(jf) = \frac{A_o f_b}{f_b + jf} \quad (2)$$

The op-amp is intentionally built with very small f_b to assure the stability of the closed loop amplifier at unity gain. Typically f_b is less than 10Hz. If the frequency of the amplified signal is chosen such that $f \gg f_b$ then equation (2) can be approximated to be

$$A(jf) \approx \frac{A_o f_b}{jf} = \frac{f_t}{jf} \quad (3)$$

Two important observations obtained from (3) are

1. When $f = A_o f_b$, the open loop gain $|A(jf)| = 1$. Hence, the unity gain bandwidth of the op-amp (f_t) is given as $f_t = A_o f_b$.
2. The gain bandwidth-product is constant. That is, at any given frequency, frequency times gain is constant such that $|jA(jf)|f \approx A_o f_b = f_t$.

In this part of the lab assignment, you will investigate the effect of the practical open loop gain $A(jf)$ on the non-inverting amplifier configuration shown in Fig. 1 below. Recall from Sedra & Smith textbook, the ideal overall gain of the non-inverting configuration is given as

$$G_{ideal} = 1 + \frac{R_2}{R_1} \quad (4)$$

By using equations (1) and (3) in the non-inverting amplifier configuration and applying KCL at the op amp's V_- input, the overall gain for the non-inverting amplifier can be shown to be:

$$G(jf) = \frac{1 + \frac{R_2}{R_1}}{1 + \frac{jf}{f_o}} = \frac{G_{ideal}}{1 + \frac{jf}{f_o}} \quad (5)$$

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In equation (5) f_o is the 3dB corner frequency of the non-inverting amplifier; it is given as

$$f_o = \frac{f_t}{1 + \frac{R_2}{R_1}}. \quad (6)$$

From equations (5) and (6), you can observe that by increasing the gain of the non-inverting amplifier its bandwidth will proportionally decrease. This leads to a very important property that is called the gain-bandwidth product such that

$$\left(1 + \frac{R_2}{R_1}\right) f_o = (G_{ideal})(f_o) \approx f_t = \text{constant}. \quad (7)$$

In this part of the lab assignment, you will experimentally validate the gain-bandwidth product for the non-inverting configuration using the OP27 op-amp. You will design three different gains and experimentally determine their corresponding 3dB corner (that is, cutoff) frequencies.

Pre-lab:

- From the OP27 datasheet, obtain the minimum and typical gain-bandwidth product.
- Refer to the non-inverting amplifier shown in Figure 1. Let $R_1=1\text{ k}\Omega$, and design an amplifier by choosing three different values for R_2 so that the gain is
 - $G_{ideal,1} \approx +11\text{ V/V}$
 - $G_{ideal,2} \approx +48\text{ V/V}$
 - $G_{ideal,2} \approx +101\text{ V/V}$
- Estimate the typical and minimum 3dB corner frequency for each of the three amplifiers based on information from (a).
- Simulate the first amplifier using PSPICE (LTSPICE or ORCAD) using the OP27 by using AC sweep and determine the 3dB frequency. Run the AC sweep from 10 Hz to 10 MHz with at least 1000 points per decade.

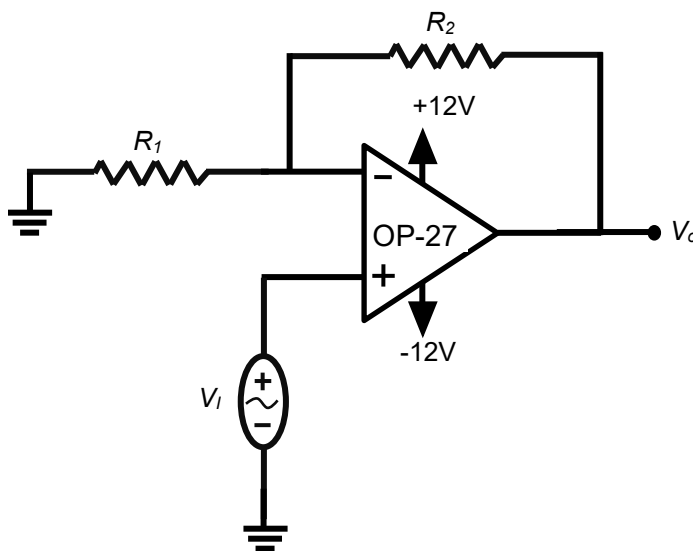


Figure 1. Non-inverting op-amp configuration.

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Lab Procedures:

1. Using the OP27 op-amp, connect the non-inverting amplifier circuit shown in Figure 1 by making $R_1=1\text{k}\Omega$, and $R_2=10\text{k}\Omega$. Record measured resistor values in your lab notebook.
2. Using a Tee connector at the function generator terminal, connect one of the Tee splitter inputs to the input of the op-amp and the other one to CH1 of the oscilloscope. Also, connect the output of the op-amp to CH2 of the scope.
3. To measure the low frequency gain of the non-inverting amplifier, apply 0.2V peak-to-peak sine wave at a low frequency 100 Hz.
4. Measure V_i and V_o and record the values on your lab notebook. Calculate the gain (V_o/V_i) and record the low frequency gain in your lab notebook. Assume this is the DC gain G_{DC} because 100 Hz is very low compared to the expected 3dB frequency.
5. To determine the 3dB corner frequency, slowly increase the frequency until the gain is $\frac{G_{DC}}{\sqrt{2}} = 0.707G_{DC}$. This is your 3dB corner frequency; record it in your lab notebook. Calculate the gain-bandwidth product and record it in your lab notebook. Compare the values with your pre-lab estimated values based on the datasheet.
6. Using Excel or Matlab, plot the frequency response of the gain by measuring the gain at different frequencies. Plot the gain in dB vs the frequency in a log scale, and collect enough points around the 3dB frequency to show a smooth curve.
7. Replace $R_2 = 47\text{ k}\Omega$, then repeat steps 2 through 5.
8. Replace $R_2 = 100\text{ k}\Omega$, then repeat steps 2 through 5.
9. *Demonstrate operation of your circuit to the Teaching Assistant for at least one of the three values of R_2 (10 k Ω , 47 k Ω , or 100 k Ω). Have the TA initial your lab check list.*

II. Slew Rate Distortion

Practical op-amps have limited maximum output current that supplies the load or sinks from the load. Also, practical op-amps have undesired output capacitance due to packaging, the internal design of the op-amp, the external layout, the breadboard or PCB and other factors. The current through the output capacitance is given as

$$I_C = C \frac{dV_o}{dt}. \quad (8)$$

Equation (8) can be rearranged as

$$\frac{dV_o}{dt} = \frac{I_C}{C}. \quad (9)$$

However, the maximum current supplied to the output capacitance is limited by the maximum output current. This limits the maximum rate of change of the output voltage which defines the slew rate (SR):

$$SR = \left. \frac{dv_o}{dt} \right|_{max} = \frac{I_{o,max}}{C}. \quad (10)$$

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Since the op-amp's output current is limited to the maximum value ($I_{o,max}$), the rate of change of the output voltage is limited by its maximum value. If the output signal is a sine wave given as

$$v_o(t) = V_m \sin(\omega t + \theta) \quad (11)$$

then its derivative is

$$\frac{dv_o}{dt} = \omega V_m \cos(\omega t + \theta). \quad (12)$$

SR determines the maximum frequency that can be used before the signal is distorted at the output which is determined by

$$SR = \left. \frac{dV_o}{dt} \right|_{max} = \omega V_m. \quad (13)$$

Hence, the maximum frequency in Hz before distorting the output signal is obtained as

$$f_{max} < \frac{SR}{2\pi V_m}. \quad (14)$$

Pre-lab:

- Determine the minimum and the typical SR for the OP27 from the datasheet. Note that the minimum SR is the worst case as it determines the slowest rate of change.
- Draw the step response of the unity gain amplifier using OP27 op-amp based on the typical value of SR and compare it to the response of the ideal op-amp. Refer to Figure 2.43 (Sedra and Smith 7th edition) on pp. 113. The unity gain amplifier is shown in Figure 2 in this handout.
- Assuming that you are applying a sine wave with amplitude $V_m = 10\text{ V}$, and based on the worst case SR of OP27, determine the maximum frequency f_{max} before the SR distortion starts taking effect.

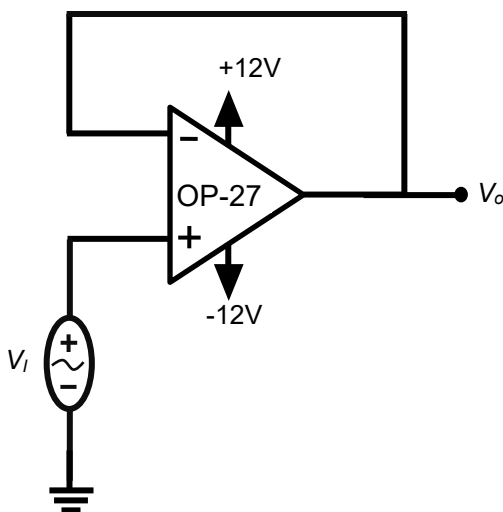


Figure 2. Unity gain amplifier using OP27 op-amp.

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Lab Procedures:

1. Using the OP27 op-amp, connect the unity gain amplifier as shown in Figure 2.
2. Using a Tee splitter and the function generator, connect one of the Tee's inputs to the op-amp and the other input to CH1 of the oscilloscope.
3. Apply a 20 V peak-to-peak square wave from the function generator. Adjust the time/div of the oscilloscope and vary the frequency until it is visually clear that the SR can be observed. Estimate the slew rate of the op-amp in the linear region by using

$$SR = \frac{\Delta V_o}{\Delta t}. \quad (15)$$

4. Compare your estimated SR value to the expected datasheet SR value. Save your graph on a thumb drive to be included in your report.
5. *Demonstrate operation of your circuit to the TA. Have the TA initial your check list.*
6. From the SR experimental value obtained in step (3), estimate f_{max} for the sine wave with amplitude $V_m = 10\text{ V}$. Record this value on your lab notebook.
7. Apply a sinewave to the input signal with 20V peak-to-peak ($V_m = 10\text{ V}$). Slowly Increase the frequency until you notice a distortion in the sine wave starts to occur. At this point, the sine wave starts to appear to be linear instead of being curvy. The beginning of the distortion appears at the 45° mark, which is location of the maximum rate of change of the sine wave. Compare the frequency at which you first notice the distortion with the estimated frequency in step (6) and with the datasheet value that is estimated in your prelab. Save your graph on a thumb drive to be included in your report.
8. *Demonstrate operation of your circuit to the TA. Have the TA initial your check list.*

III. DC Offset Voltage, DC Offset Current and DC Bias Current

Ideal op-amps require the inverting and non-inverting inputs to be perfectly matched, so that the output will be zero when the two inputs are equal. For example, the op-amp shown in Figure 3(a) has the two inputs grounded. Hence V_o should be $V_o = A(V_+ - V_-) = 0\text{V}$. In reality, the inputs are mismatched, and the output will not be zero. To model the mismatch imperfection, we assume the op-amp has an internal dc voltage at the non-inverting terminal that causes the output voltage to be non-zero as shown in Figure 3(b). Because the op-amp has very high differential gain, the output voltage will be saturated to one rail voltage. The offset input voltage is usually so small that it cannot be measured directly, so a testing circuit is used instead. The circuit shown in Figure 4 will be used to estimate the input offset voltage of the OP27 op-amp.

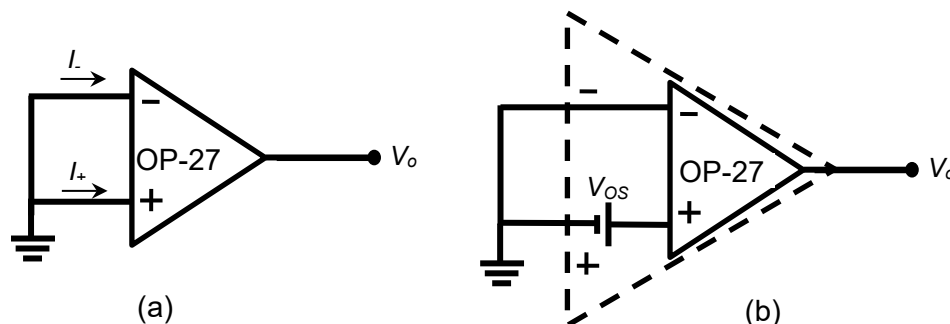


Figure 3. (a) Both inputs are grounded. (b) Modeling the mismatch as an input offset voltage.

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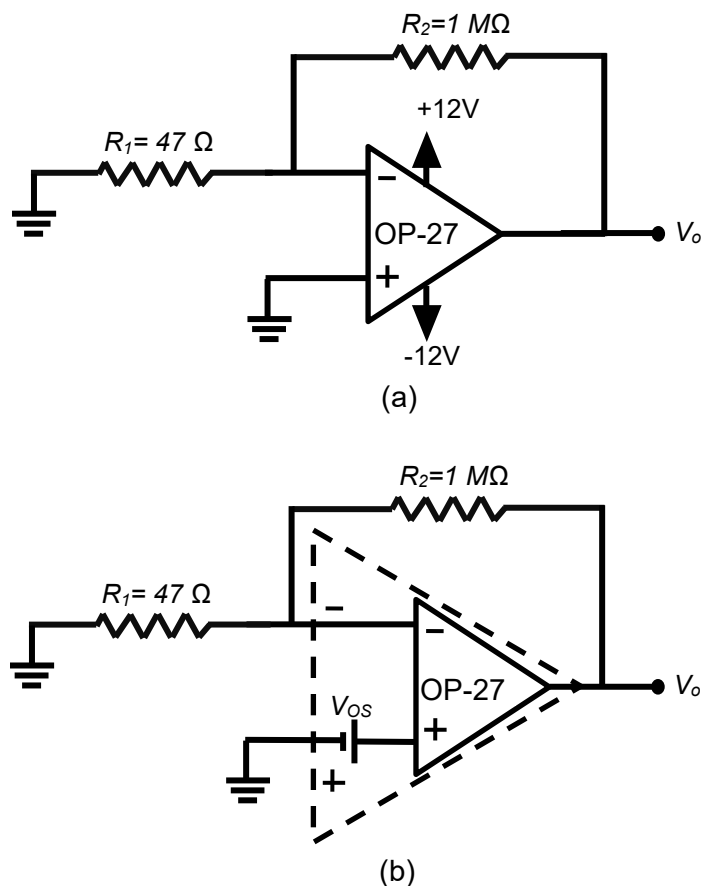


Figure 4. (a) The circuit used to estimate the input offset voltage. (b) The circuit in (a) is modeled with the input offset voltage at the non-inverting input.

Another DC imperfection of the op-amp is that the input currents to the inverting input (I_-) and the non-inverting input (I_+) are non-zero as shown in Figure 3 (a). The data sheet of a typical op-amp includes two type of DC currents, the input bias current (I_B) and the input offset current (I_{OS}), which are defined as

$$I_B = \frac{I_+ + I_-}{2} \quad (16)$$

$$I_{OS} = |I_+ - I_-|. \quad (17)$$

To evaluate I_B and I_{OS} , I_+ and I_- must be measured. These currents are usually too small to be measured directly in the lab, so testing circuits will be used to measure them. The circuit in Figure 5(a) is used to measure I_- and the circuit in Figure 5(b) is used to measure I_+ .

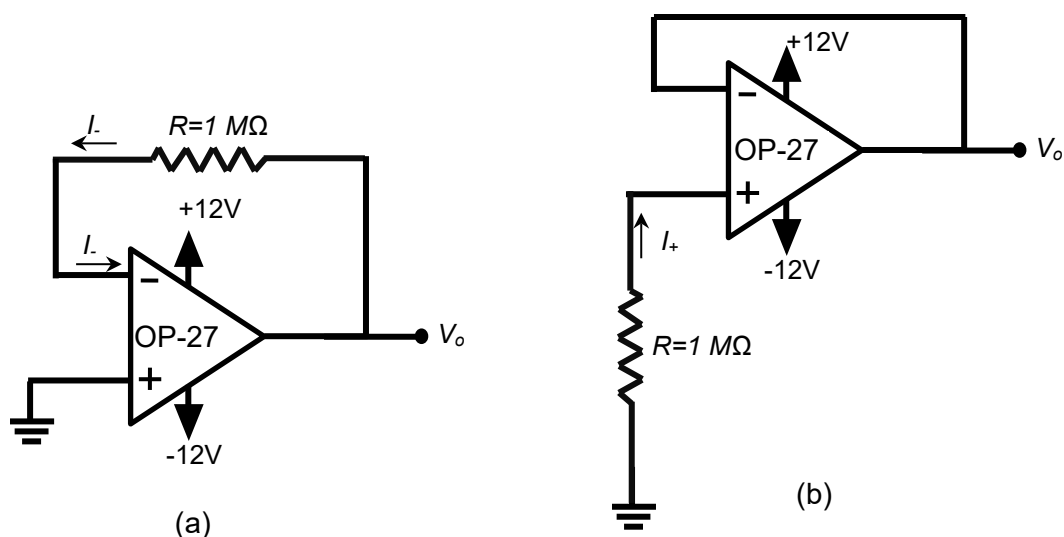


Figure 5. (a) The circuit used to estimate I_- . (b) The circuit used to estimate I_+ .

It is possible to minimize the DC imperfection of an op-amp circuit such as the one shown in Figure 6(a) by adding a compensating resistance (R_x) at the non-inverting input as shown in Figure 6(b). See Sedra and Smith for the development of the compensating resistor.

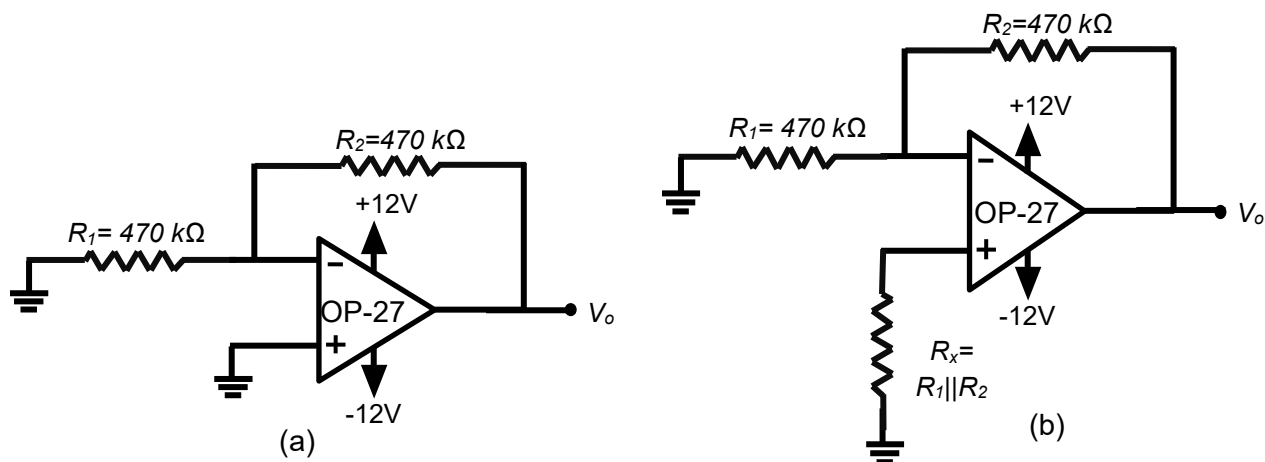


Figure 6. (a) Uncompensated circuit. (b) Compensated circuit.

Pre-lab:

- Record the typical and maximum input offset voltage, input offset current and input bias current for the OP27 op-amp from its datasheet.
- For the circuit shown in figure 4(b), estimate the typical and maximum output voltage due to the input offset voltage V_{os} .
- For the circuit in figure 5(a) and 5(b) show that the output voltage is given as

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$$I_- = \frac{V_o - V_{OS}}{R} \quad (18a)$$

$$I_+ = \frac{V_{OS} - V_o}{R}. \quad (18b)$$

Lab Procedures:

A. Estimating the input offset voltage

1. Refer to the circuit in Figure 4(a). Record the resistance values for R_1 and R_2 , and then estimate the gain according to the recorded values. Connect the circuit as shown in Figure 4(a). Measure the output voltage using the DMM.
2. Estimate the input offset voltage V_{OS} based on the measured output voltage and the estimated gain. Compare your results to the datasheet typical and maximum V_{OS} .
3. *Demonstrate operation of your circuit to the TA. Have the TA initial your check list.*

B. Estimating the input bias current and the input offset current

1. Refer to the circuit in Figure 5(a). Record the resistance value for R , and then connect the circuit as in Figure 5(a). Measure the output voltage using the DMM.
2. Estimate the inverting input current (I_-) according to (18a). Use V_{OS} from step A.
3. *Demonstrate operation of your circuit to the TA. Have the TA initial your check list.*
4. Refer to the circuit in Figure 5(b). Move the resistor R to the non-inverting input and then connect the circuit as shown. Measure the output voltage using the DMM.
5. Estimate the non-inverting input current (I_+) according to (18b). Use V_{OS} from step A.
6. Estimate the input bias current I_B and input offset current I_{OS} according to equations (16) and (17), and compare your results to the datasheet values.
7. *Demonstrate operation of your circuit to the TA. Have the TA initial your check list.*

C. DC offset compensation circuit

1. Refer to the circuit in Figure 6(a). Record the resistance values for R_1 and R_2 , and then connect the circuit as shown. Measure and record the output voltage using the DMM.
2. Add R_x to compensate for the offset measurements as shown in Figure 6(b). Measure and record the output voltage using the DMM.
3. Compare the uncompensated output voltage to the compensated output voltage.
4. *Demonstrate operation of your circuit to the TA. Have the TA initial your check list.*

IV. Common Mode Rejection Ratio

Ideal op-amps should only amplify the differential input voltage (v_d) and totally reject the common mode input voltage (v_{CM}). However, due to manufacturing limitations, imperfections and mismatches at the input stage do occur. Hence, a typical practical op-amp amplifies the

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common mode input voltage. Common mode rejection ratio (CMMR) is a metric used to quantify how good the op-amp is. CMMR is usually defined in dB and is expressed as:

$$CMMR = 20 \log_{10} \left(\frac{A_d}{A_{CM}} \right), \quad (19)$$

where A_d is the open loop differential gain of the op-amp, and A_{CM} is the open loop common mode gain of the op-amp. In this part of the lab assignment you will experimentally measure A_{CM} and A_d . Once you calculate the CMMR from your measured data, you will need to compare it with the datasheet value.

Lab Procedures:

A. Measuring the Common Mode Gain (A_{CM})

1. Using the OP27 op-amp, connect the circuit as shown in Figure 7. Apply a 10 V peak-to-peak sine wave at 200Hz from the function generator to the inverting and non-inverting inputs as shown.
2. Measure the input (v_{CM}) and output ($v_{o,CM}$) signals' peak-to-peak amplitudes using the oscilloscope. Make sure that the oscilloscope is AC coupled and adjust the time/div and the volt/div to show at least two clear signals. Save your graph on a thumb drive to be included in your report.
3. Record your peak-to-peak input and output voltages and calculate A_{CM} in your lab notebook. The common mode gain is defined as

$$A_{CM} = \frac{v_{o,CM}}{v_{CM}}$$

4. Demonstrate operation of your circuit to the TA. Have the TA initial your checklist.

B. Measuring the Differential Gain (A_d)

1. Using the OP27 op-amp, connect the circuit in Figure 8. Let $R_1=1 \text{ k}\Omega$, $R_2=4.7 \text{ k}\Omega$, $R_3=47 \text{ k}\Omega$, $R_4=47\Omega$. Measure the resistor values using the DMM and record them in your lab notebook. Use the function generator as a variable DC source by taking advantage of the DC offset. You can do that by applying the smallest (1 mV) amplitude sinewave at 100Hz, and by varying the DC offset. You will be varying the input voltage to record two measurements.
2. Adjust the DC offset until the output voltage reads $V_{o,1}= +9\text{V}$ via DMM2. Measure V_3 using DMM1 (measured voltage $V_{3,1}$), and estimate the first differential input voltage using

$$v_{d,1} = -\frac{R_4}{R_4 + R_3} V_{3,1} \approx -\frac{V_{3,1}}{1000}.$$

3. Adjust the DC offset until the output voltage reads $V_{o,2}= -9\text{V}$ via DMM2. Measure V_3 using DMM1 (measured voltage $V_{3,2}$), and estimate the second differential input voltage using

$$v_{d,2} = -\frac{R_4}{R_4 + R_3} V_{3,2} \approx -\frac{V_{3,2}}{1000}.$$

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- Now, you are ready to calculate the differential gain by taking the ratio of the difference of the two readings defined as

$$A_d = \frac{V_{o,1} - V_{o,2}}{v_{d,1} - v_{d,2}}.$$

- Demonstrate operation of your circuit to the TA. Have the TA initial your check list.*
- Based on your experimental measurements, calculate the CMMR and compare it to the OP27 datasheet.

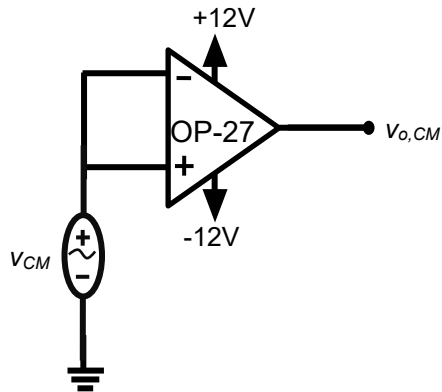


Figure 7. Common mode circuit

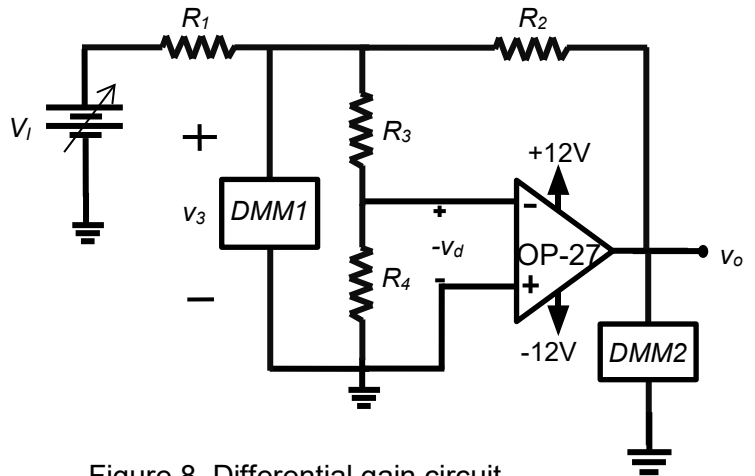


Figure 8. Differential gain circuit.

Lab 4 Checklist

Name: _____

Lab title and introduction

(5 pts)

- Lab title, your name, date, and lab partner.
- Brief introduction (one to three sentences) explaining the purpose of this lab.

I. Finite Open-Loop Bandwidth Effects (20 pts total)

1. Diagram of circuit of Figure 1. Measured values for R_1 and R_2 . (2 pts)
2. Table containing measured values for DC gains and bandwidths for $G_1=+11$ V/V, $G_2=48$ V/V and $G_3 = 101$ V/V. (4 pts)
3. Comparison of measured vs. typical and minimum gain-bandwidth product, including % error. (4 pts)
4. **DEMO:** Have a teaching assistant initial this sheet, indicating that he/she has observed your circuit's operation for at least one gain. (10 pts)
 - Circuit operates.
 - Comparison between expected typical and minimum gain-bandwidth product and measured gain-bandwidth. Show TA analysis from pre-lab providing analytical gain-bandwidth product expression.

II. Slew Rate (20 pts total)

1. Diagram of Figure 2. (2 pts)
2. Measured values for ΔV_0 and Δt for step input. Calculation of slew rate and % difference from OP27 op-amp specifications for minimum and typical SR values. Compare measured performance to specified values. (4 pts)
3. Comparison between slew rate from step response and maximum undistorted frequency from sinusoidal response. Are the two results consistent? (4 pts)
4. **DEMO:** Have a teaching assistant initial this sheet, indicating that he/she has observed your circuit's operation. (10 pts)
 - Circuit operates.
 - Give verbal assessment of comparison between specified and measured slew rate.

III. DC Imperfections (30 pts total)

A. Input Offset Voltage

1. Diagram of Figure 4(a). (1 pt)
2. Measured values for R_1 , R_2 , and V_{out} . (3 pts)
3. Show calculation of V_{os} and % difference from specifications (3 pts)
4. **DEMO:** Have a teaching assistant initial this sheet, indicating that he/she has observed your circuit's operation. (3 pts)

B. Input Bias Current and Input Offset Current

1. Diagrams for circuits 5a and 5b. (1 pt)
2. Calculated values for I_- and I_+ . (3 pts)
3. Calculation of I_B and I_{OS} and % difference from specification. (3 pts)
4. **DEMO:** Have a teaching assistant initial this sheet, indicating that he/she has observed your circuit's operation. (3 pts)

C. Offset Voltage Compensation

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1. Diagrams for circuits 6a and 6b. (1 pt)
2. Measurements for R_2 , R_1 , and V_o for Figure 6a. (3 pts)
3. Measurements for R_x and V_o for Figure 6b. (3 pts)
4. **DEMO:** Have a teaching assistant initial this sheet, indicating that he/she has observed your circuits' operation. (3 pts)

IV. Common Mode Rejection Ratio (25 pts total)

A. Common Mode Gain

1. Diagram of circuit of Figure 7. (1 pt)
2. Measurement of the common mode gain, A_{cm} . (2 pts)
3. **DEMO:** Have a teaching assistant initial this sheet, indicating that he/she has observed your circuit's operation (5 pts)

B. Differential Mode Gain

1. Diagram of circuit of Figure 8. (1 pt)
2. Measured values for R_1 , R_2 , R_3 , and R_4 . (2 pts)
3. Table of measured values for $V_{3,1}$, $V_{3,2}$, $V_{o,1}$ and $V_{o,2}$. (3 pts)
4. **DEMO:** Have a teaching assistant initial this sheet, indicating that he/she has observed your circuit's operation (5 pts)

C. Common Mode Rejection Ratio

- 1- Calculation of CMRR and % difference from specification. (4 pts)
- 2- Discuss why a high CMRR is desirable (2 pts)